

PACKING $k^2 - c$ UNIT SQUARES: A LOWER BOUND OF ORDER $k^{1/3}$ FOR THE DEFICIENCY

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ABSTRACT. Let $M(k)$ be the largest number of unit squares that can be placed in $[0, k]^2$ so that they are pairwise disjoint as closed sets; then, for integers $k \geq 2$ and $0 \leq c < k^2$, $s(k^2 - c) = k$ exactly when $c < k^2 - M(k)$. Building on [Q], where $k^2 - M(k) \geq 0.1 k^{1/4}$ is shown for every integer $k \geq 4.62 \cdot 10^{12}$, we prove that $k^2 - M(k) > 10^{-4} \sqrt{k}$ for every integer $k \in [10^8, 2.5 \cdot 10^{15}]$, that $k^2 - M(k) > 7 \cdot 10^{-5} \sqrt{k}$ for every integer $k \in [2.5 \cdot 10^{15}, 3.5 \cdot 10^{17}]$, that $k^2 - M(k) > 0.06 k^{1/3}$ for every integer $k \geq 3.5 \cdot 10^{17}$, and that $\liminf_{k \rightarrow \infty} (k^2 - M(k))/k^{1/3} \geq 0.06285$. For every $k \geq 4.62 \cdot 10^{12}$ these bounds exceed the bound of [Q]. The new ingredient is a bound for the total inclination of the squares crossed by a single path of the flow of [Q]: the height ranges of the crossings are disjoint, so the lemma of [R] on the cost of inclinations applies to their union. It allows a single flow, with a tilt threshold of order $1/W$, to keep every path quantized at every height. The constants rely on the computer-assisted Lemma 4.10 of [R]; without it the same argument gives $8.5 \cdot 10^{-5} \sqrt{k}$ on $[10^8, 2.4 \cdot 10^{15}]$, $6 \cdot 10^{-5} \sqrt{k}$ on $[2.4 \cdot 10^{15}, 3.39 \cdot 10^{17}]$, $0.05 k^{1/3}$ from $3.39 \cdot 10^{17}$ on, and the constant 0.05229. This paper has not been peer reviewed.

1. INTRODUCTION AND RESULTS

For $n \geq 1$ let $s(n)$ be the side of the smallest square that contains n unit squares with pairwise disjoint interiors. A *closed packing* of $[0, k]^2$ is a finite family of closed unit squares contained in $[0, k]^2$ that are pairwise disjoint as closed sets. Let $M(k)$ be the largest size of a closed packing of $[0, k]^2$ and $W_{\min}(k) := k^2 - M(k)$; for a given closed packing $\mathcal{P}$ we write $W := k^2 - |\mathcal{P}|$, an integer, the area of the uncovered part of $[0, k]^2$. For integers $k \geq 2$ and $0 \leq c < k^2$ one has $s(k^2 - c) = k$ if and only if $c < W_{\min}(k)$ ([Q, Section 1]).

Nagamochi stated $s(k^2 - 1) = s(k^2 - 2) = k$ for every $k \geq 2$; his published proof has a gap, and replacement proofs have been given, of $s(k^2 - 1) = k$ in a preprint and of $s(k^2 - 2) = k$ with a formalization in Lean (see [R, Section 1] for the references). Thus $W_{\min}(k) \geq 3$ for $k \geq 2$. Computer-assisted announcements that have not yet been refereed give $s(k^2 - 3) = k$ for $k \geq 6$ and $s(k^2 - 4) = k$ for $k \geq 5$, i.e. $W_{\min}(k) \geq 5$ for $k \geq 6$ ([R, Section 1]). The previous paper [R] shows $W_{\min}(k) \geq 0.0353 \log k$ for every integer $k \geq 2$, and [Q] shows $W_{\min}(k) \geq 0.1 k^{1/4}$ for every integer $k \geq 4.62 \cdot 10^{12}$ and $\liminf W_{\min}(k)/k^{1/4} \geq 0.1696$. In the other direction $W_{\min}(k) = O(k^{3/5})$, by recent work on the wasted area that has not yet been refereed ([Bui], [McC]; see [Q, Section 1]).

Theorem 1.1. *Let k be an integer and $\mathcal{P}$ a closed packing of $[0, k]^2$ with waste W .*

- (a) *If $10^8 \leq k \leq 2.5 \cdot 10^{15}$, then $W > 10^{-4} \sqrt{k}$.*
- (b) *If $2.5 \cdot 10^{15} \leq k \leq 3.5 \cdot 10^{17}$, then $W > 7 \cdot 10^{-5} \sqrt{k}$.*
- (c) *If $k \geq 3.5 \cdot 10^{17}$, then $W > 0.06 k^{1/3}$; if $k \geq 10^{21}$, then $W > 0.0625 k^{1/3}$.*
- (d) *$\liminf_{k \rightarrow \infty} W_{\min}(k)/k^{1/3} \geq 0.06285$.*

Theorem 1.2 (without [R, Lemma 4.10]). *With [R, Lemma 4.9] in place of [R, Lemma 4.10]: if $10^8 \leq k \leq 2.4 \cdot 10^{15}$, then $W > 8.5 \cdot 10^{-5} \sqrt{k}$; if $2.4 \cdot 10^{15} \leq k \leq 3.39 \cdot 10^{17}$, then $W > 6 \cdot 10^{-5} \sqrt{k}$; if $k \geq 3.39 \cdot 10^{17}$, then $W > 0.05 k^{1/3}$; if $k \geq 10^{21}$, then $W > 0.052 k^{1/3}$; and $\liminf_{k \rightarrow \infty} W_{\min}(k)/k^{1/3} \geq 0.05229$.*

Corollary 1.3. $s(k^2 - c) = k$ for every integer $k \geq 3.5 \cdot 10^{17}$ and every integer $0 \leq c \leq 0.06 k^{1/3}$, for every integer $k \in [2.5 \cdot 10^{15}, 3.5 \cdot 10^{17}]$ and every integer $0 \leq c \leq 7 \cdot 10^{-5} \sqrt{k}$, and for every integer $k \in [10^8, 2.5 \cdot 10^{15}]$ and every integer $0 \leq c \leq 10^{-4} \sqrt{k}$. Theorem 1.2 gives the same with its constants and ranges.

Proof. In each case Theorem 1.1 (resp. Theorem 1.2) gives $W_{\min}(k) > c$. $\square$

Comparison. For $k \geq 4.62 \cdot 10^{12}$ the bounds of Theorem 1.1 exceed $0.1 k^{1/4}$: on $[10^{12}, 2.5 \cdot 10^{15}]$ because $10^{-4} \sqrt{k} \geq 0.1 k^{1/4}$ there, on $[2.5 \cdot 10^{15}, 3.5 \cdot 10^{17}]$ because $7 \cdot 10^{-5} \sqrt{k} \geq 0.1 k^{1/4}$ for $k \geq 4.17 \cdot 10^{12}$, and for $k \geq 3.5 \cdot 10^{17}$ because $0.06 k^{1/3} \geq 0.1 k^{1/4}$ for $k \geq 460$. At $k = 4.62 \cdot 10^{12}$, Theorem 1.1(a) gives $W > 214.9$, so $W \geq 215$, whereas [Q] gives $W \geq 147$. As W is an integer, Theorem 1.1(a) gives $W \geq 4$ from $k \geq 9 \cdot 10^8$ on, which goes beyond $W_{\min} \geq 3$, and $W \geq 6$ from $k \geq 2.5 \cdot 10^9$ on, which goes beyond the announced $W_{\min} \geq 5$.

The constant drops from 10^{-4} in (a) to $7 \cdot 10^{-5}$ in (b) because of a trade-off in the proof (Proposition 6.1): for bounds of order $\sqrt{k}$ the range of k is limited by $W \leq W_1 \propto \tau_1$, and a larger τ_1 costs a factor $1 - 2\tau_0 < 1 - 2\tau_1$. A finer subdivision of the range would give intermediate constants.

1.1. The idea. We use the *master flow* of [Q]: from every point $(x, 0)$ of the bottom side a path moves upwards; when it meets the bottom side of a square of inclination less than a threshold θ it crosses the square along the square's own upward normal, and when it meets a square of inclination at least θ it stops (a *tilt termination*). A path that has crossed m squares $Z_1, \dots, Z_m$ arrives at height $m - D + (\text{gaps})$, where $D = \sum_j (1 - \cos \varphi_j)$ is its *deficit* ([Q, Theorem 4.2]). In [Q] the deficit is bounded by about $m \cdot \theta^2/2$ ([Q, Corollary 4.4]); to keep it small up to height s this forces $\theta \lesssim s^{-1/2}$, hence a family of flows over a continuum of scales, and the exponent $1/4$.

Here we observe that the crossings of one path occupy pairwise disjoint height intervals, and that on each of them the horizontal line meets a square of inclination $a(Z_j)$. The lemma of [R] on the cost of inclinations ([R, Lemma 4.13]) therefore applies to the union of these intervals and gives

$$\sum_j a(Z_j) \cos \varphi_j \leq (A/2 + 2\theta) W$$

for every path (Lemma 3.2). So $D \leq \frac{\theta}{2} \sum_j a(Z_j) \lesssim \theta W$, independently of the height. With $\theta \asymp 1/W$ one flow keeps all paths quantized at all heights, at the cost of tilt terminations of total measure $\lesssim W/\theta \asymp W^2$ ([Q, Theorem 8.3]). The rest of the argument is that of [Q]: the slightly tilted squares that produce the short chords required by [R, Lemma 3.5] on lines of the third kind cannot be reached by quantized paths, so they lie in the shadow of the terminated paths, whose size is bounded by [Q, Theorem 6.9]. Balancing the three kinds of lines gives $k \lesssim (kW^3)^{1/2}$, that is, $W \gtrsim k^{1/3}$.

1.2. Dependence and status. The proof uses, as black boxes, the results of [Q, Sections 3–8] listed in Section 2 and the lemmas of [R] quoted in [Q, Section 2]. Theorem 1.1 uses the computer-assisted [R, Lemma 4.10] through the constants A, A' and those of [Q, Theorem 7.1 and Proposition 5.10]; the box computations of [R, Lemma 4.10] have not yet been reproduced by an independent implementation. Theorem 1.2 does not use [R, Lemma 4.10]. The verification done so far is described at the end of the paper; none of it is by a human referee.

2. SETTING AND RESULTS USED

2.1. Conventions. We use the notation of [Q]. In particular $k \geq 4$ is an integer, $\mathcal{P}$ a closed packing of $[0, k]^2$ with waste W ; $\omega(y)$ is the length of the uncovered part of the line $\ell_y = \mathbb{R} \times \{y\}$ in $[0, k]^2$, so that $\int_0^k \omega = W$; $c_S(y)$ is the length of $S \cap \ell_y$; $\varphi_S \in (-\pi/4, \pi/4]$ is the phase of a square S and $a(S) = |\varphi_S|$ its inclination; $\text{bot}(S)$, $\text{top}(S)$ are its bottom and top sides and u_S, n_S its frame ([Q, Definition 3.7]); $\Phi(S) = \{y : 0 < c_S(y) < 1\}$; $E(y) := \sum_{S: c_S(y) \geq 1} (c_S(y) - 1)$ is the excess of the long

chords on ℓ_y ([R, Lemma 3.5]); $d(y) = \min(y, k - y)$; and $\rho(x, y) = (x, k - y)$. By [R, Corollary 3.3], $W \geq 1$ for $k \geq 2$. Throughout,

$$\delta := 10^{-5}, \quad \bar{\beta} := 1.5 \cdot 10^{-6}.$$

Constants. $A = 4\sqrt{9/0.32}/(2 - 3 \cdot 10^{-6})$ is the constant of [R, Lemmas 4.13 and 4.15] and $A' = A/\cos \bar{\beta} + 10^{-8}$ that of [Q, Theorem 8.3(a)]; $A < 10.6067 =: \bar{A}$ and $A' < 10.6068 =: \bar{A}'$. Put $S_K := 56.2501$, $C_M := 318.20$, $C_G := 56.2501$ ([Q, Theorem 7.1]) and $C_E := 754.8$ ([Q, Proposition 5.10]).

The analytic variant (without [R, Lemma 4.10]). Put $A_{13} := 4\sqrt{13/0.32}/(2 - 3 \cdot 10^{-6}) < 12.7476$ and $A'_{13} := A_{13}/\cos \bar{\beta} + 10^{-8} < 12.7476$. Then [R, Lemmas 4.13–4.15] hold with A_{13} in place of A ([Q, Remark 2.2]), and [Q, Theorem 8.3] holds with A'_{13} in place of A' ([Q, Theorem 8.3(c)]). In the analytic variant, A and A' stand for A_{13} and A'_{13} everywhere below (Lemmas 3.1, 3.2, 4.2 and 4.4, Theorem 5.1 and Section 6), and we put $\bar{A} = \bar{A}' := 12.7476$, $S_K := 81.2502$, $C_M := 459.7$, $C_G := 81.26$ and $C_E := 1090.3$ ([Q, Theorem 7.1 and Proposition 5.10]). In both cases put

$$B_1 := \frac{\bar{A}/2 + 2\bar{\beta}}{\cos \bar{\beta}}, \quad L_1 := \frac{1 + S_K \delta^2}{\delta} + (C_M + C_E + C_G) \delta + 4B_1.$$

Thus $B_1 \leq 5.3033531$ and $L_1 \leq 100021.23$, respectively $B_1 \leq 6.3738031$ and $L_1 \leq 100025.52$.

Notation. Some letters of [Q] are used here with a different meaning: H_{τ_0} (Theorem 5.1) is not the set H of [Q, Definition 3.5]; β is a number (not the function $\beta(y)$ of [Q]); $L(W)$ in Section 6 is not the function $L(y)$ of [Q]; and K denotes a set of lines, while the subscript in S_K refers to the multiplicity bound $K \in \{9, 13\}$ of [Q]. The letter E is used for $E(y)$ above, for the bound E^* of Lemma 4.2 and for the function $E(\beta, W)$ of Section 6.

2.2. The flow. Let $0 < \theta \leq \bar{\beta}$ and $k \geq 14$. Consider the parameter vector of [Q, Definition 3.1]

$$\pi_\theta := (c_0, c_1, y_0, \varepsilon, \omega_0) = (\theta, \theta/2, 1, \frac{1}{2}, \frac{1}{4}).$$

Lemma 2.1. π_θ satisfies the constraints (C0)–(C7), (Q1) and (Q2) of [Q, Definition 3.2]. Its master flow ([Q, Definition 3.20]) is $F_\theta := F(\theta, \delta, \frac{k}{2} - 1)$, with $\alpha_{\max} = \theta$.

Proof. (C0) is clear. (C1): $\alpha(y_0) = \theta \leq \bar{\beta}$. (C2): $\hat{\beta}(y_0) = \theta/2 \leq 1.5 \cdot 10^{-6}$. (C3): $c_1(1 - \varepsilon)^{-3/4} = 2^{-1/4}\theta < \theta = c_0 y_0^{1/4}$. (C4): $\frac{1}{4} < \frac{1}{2}$. (C5): with $b^* = \hat{\beta}(\frac{1}{2}) = 2^{-1/4}\theta$, $w_0 = 0.50001\theta^2(1 + 1.0001) + \delta + 1.0001 \cdot 2^{-1/4}\theta + 10^{-12} < 1.13 \cdot 10^{-5} < \frac{1}{2}$. (C6): $\delta \leq \frac{1}{2} \cos 2\theta$, which follows from (C1) by [Q, Lemma 3.3(a)]. (C7): $y_0 + 1 = 2 \leq \frac{1}{2}(\frac{k}{2} - 3)$ for $k \geq 14$. (Q1): $\theta \leq 10^{-3}$. (Q2): $b^* \leq 10^{-4}$. The master flow is $F(\alpha(y_0), \delta, \frac{k}{2} - 1)$ and $\alpha(y_0) = \theta$. $\square$

The ceiling version $\tilde{F}_\theta$ of F_θ is the master flow of $\rho\mathcal{P}$ transported by ρ ([Q, Definition 3.24]); $\rho\mathcal{P}$ is a closed packing with the same W ([Q, Lemma 3.8]). Only the master flow is used; the scale flows of [Q] play no role.

Choosing θ from W . The parameter θ will be chosen depending on W (Definition 3.4; $W \geq 1$, so θ is well defined). This is legitimate: every result of [Q] used below is a statement about one fixed closed packing and one parameter vector, and its hypotheses do not involve the packing. Precisely, Proposition 5.10 and Theorem 7.1 assume the standing assumptions of [Q, Section 5.1] ($0 < \delta \leq 10^{-5}$, $0 < \alpha_{\max} \leq 1.5 \cdot 10^{-6}$, $h_{\max} = \frac{k}{2} - 1$); Theorem 8.3 assumes $0 < \delta \leq 10^{-5}$, (C0) and (C1); Theorem 6.9 is stated for the parameters of [Q, Section 3], and, besides the definitions of the flows and [Q, Lemma 6.1], its proofs use only [Q, (6.1)] ([Q, Remark 6.13]); the results of [Q, Sections 3 and 4] used below hold for any flow $F(\alpha_F, \delta, h_F)$ or under (C0), (C1). Since $\rho\mathcal{P}$ has the same W , the floor and the ceiling flow have the same θ ; this matters because Proposition 5.10 and Theorems 7.1 and 8.3 are joint statements about the floor and the ceiling master flow of one parameter vector.

2.3. Results used. From [Q] (for the master flow F_θ , and by reflection for $\tilde{F}_\theta$):

- Remark 2.2 (the analytic variant: [R, Lemmas 4.13–4.15] with A_{13});
- Definitions 3.1 and 3.2 (parameters, constraints) and Lemma 3.3(a); Lemma 3.8 (reflection of the packing);
- Definition 3.9 (the flow; in particular $t_D = \delta - g$, so the cumulative gap never exceeds δ), Definition 3.10 (rule R2: a square is passed only if its inclination is $< \alpha_{\max}$), Definition 3.14 (rule R1), Remark 3.34 (rule R1 is applied before rule R2), Lemma 3.17(a),(e),(f) (trajectories);
- Definitions 3.20 and 3.24 (master and ceiling flows); Definitions 3.26, 3.28 and 3.30 (losses L_* , line quantities μ_Z , Sh, and Λ), Lemma 3.29(a) (crossings), Lemma 3.31 (measurability), Definitions 3.32 and 6.6 and Lemma 3.33(a) (the set Exc is finite);
- Theorem 4.2 and Corollary 4.4 (height identity), Lemma 4.5(a) (bottom side), Lemma 4.6(c) (auxiliary inequality), Lemma 4.7(a)–(c) (reflection);
- Proposition 5.10 (end collisions), Theorem 6.9(f) (shadow inequality) with Remark 6.13 and (6.1), Theorem 7.1 (deaths, merges, gap overlaps), Theorem 8.3(a),(c) (tilt terminations).

From [R] (as quoted in [Q, Section 2]): Lemma 3.1(a),(b) (vertical extent and ramp set $\mathcal{R}(S)$), Lemma 3.4 (second part: $\Phi(S) \subset \mathcal{R}(S)$ and $\int_{\Phi(S)} c_S = \sin a \cos a$ for $a = a(S) > 0$, $\Phi(S) = \emptyset$ for $a(S) = 0$), Lemma 3.5 (short chords and the excess $E(y)$ of the long chords), Corollary 3.3 ($W \geq 1$), Lemma 4.13 and Lemma 4.15.

3. THE TILT BUDGET

Lemma 3.1 (tilt budget). *Let $0 < \theta \leq \bar{\beta}$, let $I_1, \dots, I_m \subset (0, k)$ be pairwise disjoint intervals and $Z_1, \dots, Z_m \in \mathcal{P}$ (not necessarily distinct) such that $a(Z_j) < \theta$ and ℓ_y meets Z_j for every $y \in I_j$. Then*

$$\sum_{j=1}^m a(Z_j) |I_j| \leq \left(\frac{A}{2} + 2\theta \right) W.$$

Proof. Put $Y := \bigcup_j I_j$ and $\beta(y) := a(Z_j)$ for $y \in I_j$, a Borel function with values in $[0, \theta]$. Let $Y_1 := \{y \in Y : \omega(y) < \frac{1}{2}\}$. For $y \in Y_1$ the line y meets a square of inclination $\geq \beta(y)$, namely Z_j , and $\omega(y) < \frac{1}{2}$; so [R, Lemma 4.13], applied with θ in the role of its constant $\bar{\beta}$, gives $\int_{Y_1} 2\beta \leq AW$. Since $\int_0^k \omega = W$, the set $\{\omega \geq \frac{1}{2}\}$ has measure $\leq 2W$, so $\int_{Y \setminus Y_1} \beta \leq 2\theta W$. Adding, $\sum_j a(Z_j) |I_j| = \int_Y \beta \leq AW/2 + 2\theta W$. $\square$

[R, Lemma 4.13] is stated for an arbitrary (measurable) set $Y \subset [0, k]$ and function β with the stated properties; here Y is a finite union of intervals and β a step function. Its constant A does not depend on the value of $\bar{\beta} \leq 1.5 \cdot 10^{-6}$: the proof uses $\bar{\beta}$ only through the cap $\theta_{\max} = 2\bar{\beta} \leq 3 \cdot 10^{-6}$.

Lemma 3.2 (one path). *Let $0 < \theta \leq \bar{\beta}$ and let $Z_1, \dots, Z_m$ be the squares passed by a path of F_θ , in order, with $\varphi_j := \varphi_{Z_j}$. Then*

$$\sum_{j=1}^m a(Z_j) \cos \varphi_j \leq \left(\frac{A}{2} + 2\theta \right) W \quad \text{and} \quad \sum_{j=1}^m a(Z_j) \leq B_1 W.$$

The same holds for the paths of $\tilde{F}_\theta$.

Proof. Let q_j be the entry point into Z_j ; the pass segment is $[q_j, q_j + n_{Z_j}]$ and its height range is $I_j := (q_{j,y}, q_{j,y} + \cos \varphi_j)$. Its open part lies in $\text{int } Z_j$ ([Q, Lemma 3.17(e)]), so ℓ_y meets Z_j for $y \in I_j$. Heights increase strictly along a trajectory ([Q, Lemma 3.17(a)]), so the I_j are pairwise disjoint. By rule R2 ([Q, Definition 3.10]) $a(Z_j) < \theta$. Lemma 3.1 gives the first inequality; the second follows from $\cos \varphi_j \geq \cos \bar{\beta}$ and $A \leq \bar{A}$. For $\tilde{F}_\theta$ apply this to $\rho\mathcal{P}$. $\square$

Lemma 3.3 (lateral drift). *Let $0 < \theta \leq \bar{\beta}$. Every point p of the trajectory of the path of F_θ that starts at $(x, 0)$ satisfies $|p_x - x| \leq B_1 W + \delta$. Consequently the set of x whose path terminates at a side wall (type W) has measure $L_W \leq 2(B_1 W + \delta)$, and the same holds for $\tilde{F}_\theta$.*

Proof. A pass through Z_j moves the path horizontally by $|\sin \varphi_j| \leq a(Z_j)$. The gap segments have total length at most δ at every point of the trajectory: by [Q, Definition 3.9] the cumulative gap g is the total length of the gap segments travelled, and $t^* \leq t_D = \delta - g$ in every ray step. Hence $|p_x - x| \leq \sum_j a(Z_j) + \delta \leq B_1 W + \delta$ by Lemma 3.2. A W -termination point has abscissa 0 or k , so $\min(x, k - x) \leq B_1 W + \delta$. $\square$

Definition 3.4. For $\delta < \tau_1 < \frac{1}{2}$ put $\theta := \min(\bar{\beta}, 2\tau_1/(B_1 W))$ and let $F_\theta, \tilde{F}_\theta$ be the flows of Section 2 for this θ . From here on $k \geq 14$, as required by Lemma 2.1.

Lemma 3.5 (quantization). *Let θ be as in Definition 3.4. Let q be the point of an entry candidate of a path of F_θ , reached after passing $Z_1, \dots, Z_m$. Then*

$$q_y - m \in [-D, \delta), \quad D := \sum_{j=1}^m (1 - \cos \varphi_j) \leq \tau_1,$$

so $\text{dist}(q_y, \mathbb{Z}) \leq \tau_1$. For $\tilde{F}_\theta$ the same holds with $k - q_y$ in place of q_y .

Proof. By [Q, Theorem 4.2 and Corollary 4.4], $q_y - m = -D + \sum_i g_i \cos \varphi_i$, and the last sum lies in $[0, \delta)$ at an entry candidate, since the cumulative gap there is $< \delta$ ([Q, Lemma 3.17(f)]). Since $|\varphi_j| = a(Z_j) < \theta$, $1 - \cos \varphi_j \leq \varphi_j^2/2 \leq \frac{\theta}{2} a(Z_j)$. The squares passed before q are passed by the path, so Lemma 3.2 gives $D \leq \frac{\theta}{2} B_1 W \leq \tau_1$. As $\delta < \tau_1$, $\text{dist}(q_y, \mathbb{Z}) \leq \max(D, \delta) \leq \tau_1$. $\square$

4. UNREACHABLE SQUARES AND THE SHADOW

Lemma 4.1 (unreachable squares). *Let θ be as in Definition 3.4, $\tau_1 < \tau_0 < \frac{1}{2}$ and $0 < \beta \leq \bar{\beta}$ with $\tau_0 - 1.0001\beta > \tau_1$. Let $S \in \mathcal{P}$ with $0 < a(S) < \beta$, and suppose that $\Phi(S)$ contains a height y with $\text{dist}(y, \mathbb{Z}) \geq \tau_0$. Then no path of F_θ has an entry candidate on S , and $\mu_S^{F_\theta}(y') = 0$ for every $y' \in (0, \frac{k}{2} - 1)$. Likewise no path of $\tilde{F}_\theta$ has an entry candidate on $\text{top}(S) = \rho(\text{bot}(\rho S))$, and $\mu_S^{\tilde{F}_\theta} \equiv 0$.*

Proof. Let $a := a(S)$ and let v be the lowest height of S . By [R, Lemmas 3.4 and 3.1(b)], $y \in \Phi(S) \subset \mathcal{R}(S) = [v, v + \sin a] \cup (v + \cos a, v + \cos a + \sin a]$, and the heights of the points of $\text{bot}(S)$ fill exactly $[v, v + \sin a]$ ([Q, Lemma 4.5(a)]). If $y < v + \sin a$, every $z \in \text{bot}(S)$ has $|z_y - y| \leq \sin a < \beta$. If $y > v + \cos a$, every $z \in \text{bot}(S)$ has $|z_y - (y - 1)| \leq (1 - \cos a) + \sin a \leq 1.0001a < 1.0001\beta$ ([Q, Lemma 4.6(c)], as $a < \bar{\beta} \leq 2 \cdot 10^{-4}$). In both cases $\text{dist}(z_y, \mathbb{Z}) \geq \tau_0 - 1.0001\beta > \tau_1$, which contradicts Lemma 3.5 for an entry candidate at z . Entry candidates lie in $\text{relint bot}(S)$ ([Q, Definition 3.9]); a path crosses $\ell_{y'}$ in $\text{int } S$ only if it passes S ([Q, Lemma 3.29(a)]), and passing requires an entry candidate on S . Hence $\mu_S^{F_\theta} \equiv 0$. The ceiling statement is the floor statement for $\rho\mathcal{P}$, using $a(\rho S) = a(S)$, $\rho(\text{bot}(\rho S)) = \text{top}(S)$, $\Phi(\rho S) = k - \Phi(S)$ and $\text{dist}(k - y, \mathbb{Z}) = \text{dist}(y, \mathbb{Z})$ (k is an integer) ([Q, Lemma 4.7(a)–(c)]). $\square$

Lemma 4.2 (short chords). *Let $0 < \beta \leq \bar{\beta}$, $0 < \nu_0 < \frac{1}{2}$, and let $K \subset (0, k)$ be a Borel set such that for every $y \in K$: $d(y) \leq \frac{k}{2} - 1$, $\omega(y) < \nu_0$, and every square meeting ℓ_y has inclination $< \beta$. Let $\mathcal{Z}_K := \{S \in \mathcal{P} : \Phi(S) \cap K \neq \emptyset\}$. Then*

$$\beta |\mathcal{Z}_K| \geq (1 - \nu_0 - E^*) |K|, \quad E^* := 0.50001 \beta \left(A + \frac{\beta}{\delta} \right) W.$$

Proof. For $y \in K$, [R, Lemma 3.5] gives $\sum_{S: 0 < c_S(y) < 1} c_S(y) \geq 1 - \omega(y) - E(y)$, and every S in this sum has $y \in \Phi(S)$, so $S \in \mathcal{Z}_K$. By [R, Lemma 4.15] (with $\alpha := \beta$) $E(y) \leq E^*$ on K . Each $S \in \mathcal{Z}_K$ meets a line of K , so $a(S) < \beta$, and $a(S) > 0$ as $\Phi(S) \neq \emptyset$; by [R, Lemma 3.4], $\int_{\Phi(S)} c_S = \sin a \cos a < \beta$. Integrating over K (the sum is finite), $(1 - \nu_0 - E^*) |K| \leq \sum_{S \in \mathcal{Z}_K} \int_{K \cap \Phi(S)} c_S \leq \beta |\mathcal{Z}_K|$. $\square$

Lemma 4.3 (shadow budget). *Let θ be as in Definition 3.4, $h := \frac{k}{2} - 1$, and let $\mathcal{Z}$ be a set of squares $S \in \mathcal{P}$ with $S \subset \mathbb{R} \times [0, h]$ and $\mu_S^{F_\theta} \equiv 0$ on $(0, h)$. Then $|\mathcal{Z}| \leq h(L_T + \Lambda_b)$, where L_T is the measure of the set of floor points whose path in F_θ is T-terminated, L_W that of the W-terminated ones (Lemma 3.3), and $\Lambda_b := |\mathcal{D}| + |\mathcal{M}| + L_E + L_W + \sup_{0 < y < h} \text{Ov}_{\text{gap}}^{F_\theta}(y)$. The same holds for $\tilde{F}_\theta$, for squares $S \in \mathcal{P}$ with $S \subset \mathbb{R} \times [k - h, k]$ and $\mu_S^{\tilde{F}_\theta} \equiv 0$ on $(k - h, k)$.*

Proof. Let Exc be the exceptional set of the master flow ([Q, Definitions 3.32 and 6.6]); it is finite ([Q, Lemma 3.33(a)]). For $y \in (0, h) \setminus \text{Exc}$, [Q, Theorem 6.9(f)] gives $\text{Sh}(y) \leq L_T(y) + \Lambda(y) \leq L_T + \Lambda_b$, since each $L_*(y)$ counts only terminations below y and $\text{Ov}_{\text{gap}}(y) \leq \sup_{0 < y' < h} \text{Ov}_{\text{gap}}(y')$ ([Q, Definitions 3.26 and 3.30]). By [Q, Definition 3.28], $\text{Sh}(y) = \sum_Z (c_Z(y) - \mu_Z(y))_+ \geq \sum_{S \in \mathcal{Z}} c_S(y)$. As Sh is Borel ([Q, Lemma 3.31(d)]), integrating over $(0, h)$ gives $\sum_{S \in \mathcal{Z}} \int_0^h c_S = |\mathcal{Z}| \leq h(L_T + \Lambda_b)$. $\square$

Lemma 4.4 (terminations). *Let θ be as in Definition 3.4 and let L_{tot} be the sum of $L_T + \Lambda_b$ for F_θ and for $\tilde{F}_\theta$. Then*

$$L_{\text{tot}} \leq \frac{\bar{A}'W}{\theta} + L_1W + 4\delta = \bar{A}'W \max\left(\frac{1}{\bar{\beta}}, \frac{B_1W}{2\tau_1}\right) + L_1W + 4\delta.$$

Proof. A path that is T-terminated in F_θ is the R1 winner at its last entry ([Q, Definition 3.14]; rule R1 is applied before rule R2, [Q, Remark 3.34]), at a square of inclination $\geq \alpha_{\max} = \theta$. So [Q, Theorem 8.3(a)] (resp. (c)), with $\mathcal{X}$ and $\mathcal{X}'$ the T-terminated floor and ceiling points (finite unions of intervals and points, [Q, Lemma 3.31(a)]) and $\zeta \equiv \zeta' \equiv \theta \leq \bar{\beta}$, gives $\theta(L_T + \tilde{L}_T) \leq \bar{A}'W$. By [Q, Theorem 7.1], $|\mathcal{D}| + |\tilde{\mathcal{D}}| \leq (1 + S_K\delta^2)W/\delta$, $|\mathcal{M}| + |\tilde{\mathcal{M}}| \leq C_M\delta W$ and the two suprema of Ov_{gap} add up to at most $C_G\delta W$; by [Q, Proposition 5.10], $L_E + \tilde{L}_E \leq C_E\delta W$; and by Lemma 3.3 the two wall terms add up to at most $4(B_1W + \delta)$. (The hypotheses of these results of [Q] are $0 < \delta \leq 10^{-5}$, $0 < \alpha_{\max} \leq 1.5 \cdot 10^{-6}$, $h_{\max} = \frac{k}{2} - 1$, and (C0), (C1); they hold by Lemma 2.1.) $\square$

5. THE MAIN INEQUALITY

Theorem 5.1. *Let $k \geq 14$ be an integer and $\mathcal{P}$ a closed packing of $[0, k]^2$ with waste W . Let $\delta < \tau_1 < \tau_0 < \frac{1}{2}$, $0 < \nu_0 < \frac{1}{2}$ and $0 < \beta \leq \bar{\beta}$ with $\tau_0 - 1.0001\beta > \tau_1$ and $E^* = 0.50001\beta(A + \beta/\delta)W < 1 - \nu_0$, and let θ be as in Definition 3.4 for this τ_1 . Then*

$$(5.1) \quad (1 - 2\tau_0)(k - 6) \leq \frac{W}{\nu_0} + \frac{AW}{2\beta} + \frac{\beta(\frac{k}{2} - 1)L_{\text{tot}}}{1 - \nu_0 - E^*},$$

with L_{tot} bounded by Lemma 4.4.

Proof. Let $H_{\tau_0} := \{y : \text{dist}(y, \mathbb{Z}) \geq \tau_0\}$ and $\mathcal{H} := H_{\tau_0} \cap ((0, \frac{k}{2} - 3) \cup (\frac{k}{2} + 3, k))$. As k is an integer, $(0, \frac{k}{2} - 3)$ consists of $\frac{k}{2} - 3$ unit intervals $[n, n + 1)$, or of $\frac{k-7}{2}$ unit intervals and one half interval $[n, n + \frac{1}{2})$; H_{τ_0} has measure $1 - 2\tau_0$ in each unit interval and $\frac{1}{2} - \tau_0$ in each such half interval; the same holds for $(\frac{k}{2} + 3, k)$. So $|\mathcal{H}| = (1 - 2\tau_0)(k - 6)$. Split $\mathcal{H}$ into $J_1 := \{y \in \mathcal{H} : \omega(y) \geq \nu_0\}$, $J_2 := \{y \in \mathcal{H} : \omega(y) < \nu_0 \text{ and } \ell_y \text{ meets a square of inclination } \geq \beta\}$, and $K := \mathcal{H} \setminus (J_1 \cup J_2)$ (lines of the first, second and third kind).

(i) $|J_1| \leq W/\nu_0$, since $\int \omega = W$.

(ii) $|J_2| \leq AW/(2\beta)$: apply [R, Lemma 4.13] with β as its constant and $\beta(\cdot) \equiv \beta$ on J_2 , a finite union of intervals; $\omega < \nu_0 < \frac{1}{2}$ there.

(iii) Let $K_b := K \cap (0, \frac{k}{2} - 3)$ and $K_t := K \cap (\frac{k}{2} + 3, k)$. On K_b , $d(y) = y < \frac{k}{2} - 3$, so Lemma 4.2 applies to K_b . Every $S \in \mathcal{Z}_{K_b}$ satisfies the hypotheses of Lemma 4.1 ($0 < a(S) < \beta$, and $\Phi(S)$ meets $K_b \subset H_{\tau_0}$), so $\mu_S^{F_\theta} \equiv 0$; moreover S meets a line of K_b and has vertical extent $\cos a + \sin a < 1.0001$ ([R, Lemma 3.1(a)]), so $S \subset \mathbb{R} \times [0, \frac{k}{2} - 1.9999] \subset \mathbb{R} \times [0, h]$. By Lemmas 4.2 and 4.3, $(1 - \nu_0 - E^*)|K_b| \leq \beta|\mathcal{Z}_{K_b}| \leq \beta h(L_T + \Lambda_b)$. The same argument with $\tilde{F}_\theta$ gives $(1 - \nu_0 - E^*)|K_t| \leq \beta h(\tilde{L}_T + \tilde{\Lambda}_b)$. Adding, $|K| \leq \beta h L_{\text{tot}}/(1 - \nu_0 - E^*)$.

Summing (i)–(iii) gives (5.1). $\square$

6. PROOF OF THE EXPLICIT BOUNDS

Write $L(W) := \bar{A}'W \max(1/\bar{\beta}, B_1 W/(2\tau_1)) + L_1 W + 4\delta$ and $E(\beta, W) := 0.50001\beta(\bar{A} + \beta/\delta)W$. Since $A \leq \bar{A}$, Theorem 5.1 remains true with A replaced by $\bar{A}$, E^* by $E(\beta, W)$ and L_{tot} by $L(W)$.

Proposition 6.1. *Let $p \in \{\frac{1}{2}, \frac{1}{3}\}$, $\gamma, b > 0$, $\delta < \tau_1 < \tau_0 < \frac{1}{2}$, $0 < \nu_0 < \frac{1}{2}$ and $14 \leq k_a \leq k_b \leq \infty$. Put $\beta(k) := bk^{p-1}$, $W(k) := \gamma k^p$ and*

$$R^+(k) := \frac{W(k)}{\nu_0} + \frac{\bar{A}W(k)}{2\beta(k)} + \frac{\beta(k)^{\frac{k}{2}} L(W(k))}{1 - \nu_0 - E(\beta(k), W(k))}.$$

Assume: (i) $\beta(k_a) \leq \bar{\beta}$ and $\tau_0 - 1.0001\beta(k_a) > \tau_1$; (ii) $E(\beta(k_a), W(k_a)) < 1 - \nu_0$; (iii) $(1 - 2\tau_0)(k_a - 6) > R^+(k_a)$; (iv) if $p = \frac{1}{2}$, then $\gamma k_b^{1/2} \leq W_1 := 2\tau_1/(B_1\bar{\beta})$. Then every closed packing of $[0, k]^2$ with an integer $k \in [k_a, k_b]$ has $W > \gamma k^p$.

Proof. For $k \geq k_a$ the functions $\beta(k)$ and $E(\beta(k), W(k)) = 0.50001 b\gamma k^{2p-1}(\bar{A} + bk^{p-1}/\delta)$ are non-increasing, so (i) and (ii) hold at every $k \in [k_a, k_b]$. Next, $R^+(k)/k$ is non-increasing on $[k_a, k_b]$: its terms are $\gamma k^{p-1}/\nu_0$, the constant $\bar{A}\gamma/(2b)$, and $\beta L/(2(1 - \nu_0 - E))$, where $\beta \bar{A}'W \max(\dots) = \max(b\gamma \bar{A}'k^{2p-1}/\bar{\beta}, b\gamma^2 \bar{A}'B_1 k^{3p-1}/(2\tau_1))$, $\beta L_1 W = b\gamma L_1 k^{2p-1}$ and $4\delta\beta$. For $p = \frac{1}{3}$ all exponents are ≤ 0 . For $p = \frac{1}{2}$, (iv) gives $W(k) \leq W_1$, so the maximum is $1/\bar{\beta}$ and the remaining exponents are ≤ 0 . On the other hand $(1 - 2\tau_0)(k - 6)/k$ is increasing. By (iii), $(1 - 2\tau_0)(k - 6) > R^+(k)$ for all $k \in [k_a, k_b]$.

Now let $\mathcal{P}$ be a closed packing of $[0, k]^2$, $k \in [k_a, k_b]$ an integer, with $W \leq W(k)$. Theorem 5.1 applies with $\beta = \beta(k)$: $E(\beta, W) \leq E(\beta, W(k)) < 1 - \nu_0$. Its right side is non-decreasing in W (as $\theta(W)$ is non-increasing, $W/\theta(W)$ and E are increasing). Hence $(1 - 2\tau_0)(k - 6) \leq R^+(k)$, a contradiction. $\square$

case	p	γ	b	$[k_a, k_b]$	τ_0	τ_1	ν_0	left/ $k_a \geq$	$R^+(k_a)/k_a \leq$
1	$\frac{1}{2}$	10^{-4}	$1.2 \cdot 10^{-3}$	$[10^8, 2.5 \cdot 10^{15}]$	0.0205	0.02	0.001	0.95899994	0.87266009
2	$\frac{1}{2}$	$7 \cdot 10^{-5}$	$1.2 \cdot 10^{-3}$	$[2.5 \cdot 10^{15}, 3.5 \cdot 10^{17}]$	0.1675	0.167	0.001	0.66499999	0.61085501
3	$\frac{1}{3}$	0.06	1	$[3.5 \cdot 10^{17}, \infty)$	0.1675	0.167	0.01	0.66499999	0.62871651
4	$\frac{1}{3}$	0.0625	1	$[10^{21}, \infty)$	0.1668	0.1667	0.005	0.66639999	0.66296281
5	$\frac{1}{2}$	$8.5 \cdot 10^{-5}$	$1.2 \cdot 10^{-3}$	$[10^8, 2.4 \cdot 10^{15}]$	0.0205	0.02	0.001	0.95899994	0.89044495
6	$\frac{1}{2}$	$6 \cdot 10^{-5}$	$1.2 \cdot 10^{-3}$	$[2.4 \cdot 10^{15}, 3.39 \cdot 10^{17}]$	0.1675	0.167	0.001	0.66499999	0.62854332
7	$\frac{1}{3}$	0.05	1	$[3.39 \cdot 10^{17}, \infty)$	0.1675	0.167	0.01	0.66499999	0.63009391
8	$\frac{1}{3}$	0.052	1	$[10^{21}, \infty)$	0.1668	0.1667	0.005	0.66639999	0.66284161

TABLE 1. The eight cases of Proposition 6.1: cases 1–4 with the constants that use [R, Lemma 4.10], cases 5–8 with the analytic constants. The last two columns are a lower bound for $(1 - 2\tau_0)(k_a - 6)/k_a$ and an upper bound for $R^+(k_a)/k_a$, rounded outwards to eight digits from the interval computation. R^+ is computed with $\bar{A}$, $\bar{A}'$, S_K , C_M , C_G , C_E of Section 2 and with B_1 , L_1 as defined there. With the decimal bounds for B_1 and L_1 in their place, $R^+(k_a)/k_a$ exceeds the entries of cases 3 and 4 by about $5 \cdot 10^{-9}$, and (iii) still holds.

Proof of Theorem 1.1(a)–(c) and of the explicit bounds of Theorem 1.2. Apply Proposition 6.1 to the cases of Table 1. Conditions (i)–(iv) were checked in interval arithmetic by three programs in the folder `code/` that accompanies this paper. The first, `check_table_iv.py`, uses mpmath interval arithmetic with 40 digits. The other two, `check_table_exact.py` and `check_table_exact2.py`, use exact rational arithmetic with rigorous bounds for roots, sine and cosine; they were written from the statements, independently of the first program and of each other. In all cases

$\beta(k_a) \leq 2.5 \cdot 10^{-7}$ and $E(\beta(k_a), W(k_a)) \leq 6.6 \cdot 10^{-7}$. The smallest margin in (iv) occurs in case 6, where $\gamma k_b^{1/2} \leq 34934.23$ and $W_1 \geq 34934.66$. $\square$

Proof of the lim inf statements. Let $\bar{c}_\infty := ((4/27)/(\bar{A}\bar{A}'B_1))^{1/3}$; then $\bar{c}_\infty > 0.062853$, resp. > 0.052297 . Let $0 < \gamma < \bar{c}_\infty$. Since $\max_{0 < \tau < 1/2} \tau(1-2\tau)^2 = \frac{2}{27}$, attained at $\tau = \frac{1}{6}$, we can choose $\delta < \tau_1 < \tau_0 < \frac{1}{2}$ close to $\frac{1}{6}$ and $\nu_0 > 0$ small with $\gamma^3 < 2\tau_1(1-\nu_0)(1-2\tau_0)^2/(\bar{A}\bar{A}'B_1)$. Put $p = \frac{1}{3}$ and $b := (2\bar{A}\tau_1(1-\nu_0)/(\bar{A}'B_1\gamma))^{1/2}$. Then

$$\frac{\bar{A}\gamma}{2b} + \frac{\bar{A}'B_1b\gamma^2}{4\tau_1(1-\nu_0)} = \gamma^{3/2} \left(\frac{\bar{A}\bar{A}'B_1}{2\tau_1(1-\nu_0)} \right)^{1/2} < 1 - 2\tau_0.$$

As $k \rightarrow \infty$, $R^+(k)/k$ tends to the left side of this inequality, $\beta(k) \rightarrow 0$ and $E \rightarrow 0$, while $(1-2\tau_0)(k-6)/k \rightarrow 1-2\tau_0$. Hence (i)–(iii) hold for some finite k_a , and Proposition 6.1 gives $W_{\min}(k) > \gamma k^{1/3}$ for all $k \geq k_a$. $\square$

7. REMARKS

Remark 7.1 (where the gain comes from). There are two new estimates. The first is Lemma 3.1, which is used through Lemma 3.2; the second is Lemma 3.3, which replaces the wall term $2(s+2)\tan\alpha(s)$ of the scale flows of [Q]. In [Q] the deficit of a path at height q_y is bounded by $(\sec\alpha_{\max}-1)q_y$ ([Q, Corollary 4.4]), which grows with the height. Lemma 3.2 bounds it by $\frac{\theta}{2}B_1W$, independently of the height.

The threshold of Definition 3.4 equals $\bar{\beta}$ exactly when $W \leq W_1 := 2\tau_1/(B_1\bar{\beta})$. With the constants of Theorem 1.1, $W_1 \approx 5.03 \cdot 10^3$ for $\tau_1 = 0.02$, and $W_1 \approx 4.20 \cdot 10^4$ for $\tau_1 = 0.167$ (with the analytic constants, $W_1 \approx 4183.79$ and 34934.66). In cases 1, 2, 5 and 6 of Table 1, condition (iv) keeps W in this range. The flow is then the master flow of [Q] with $\alpha_{\max} = \bar{\beta}$, and the bounds are of order $\sqrt{k}$.

Remark 7.2 (the exponent $1/3$; heuristic). In (5.1) the term $\bar{A}'W/\theta \asymp W^2$ of L_{tot} comes from tilt terminations. Heuristically, a row of about W^2 squares of inclination about $1/W$, each sitting on axis-parallel squares, costs waste of order W and stops paths of total measure of order W^2 when $\theta \asymp 1/W$. This suggests that the bound of [Q, Theorem 8.3] used in Lemma 4.4 is attained up to a constant factor. Another source of loss is the factor $h = \frac{k}{2} - 1$ in Lemma 4.3, which charges every terminated path for the whole height. An exponent larger than $1/3$ would need a different treatment of these terms. We do not claim more.

VERIFICATION STATUS

This paper has not been peer reviewed, and neither have [Q] and [R]. So far it has been checked only by AI systems and by computer programs.

- The argument was checked by two independent blank-slate AI reviews, of a first and of a corrected draft, and a draft of this text by a third one. Each read the cited statements and, where generality matters, the proofs in [Q] and [R] (in particular the proof of [R, Lemma 4.13] for arbitrary Y and β). None of them found a critical or a major issue; their minor remarks (12, 6 and 12 of them) have been addressed.
- A further blank-slate AI review of the version of this text prepared for release was made in a separate session by a lead reviewer and five independent agents with separate scopes ([Q, Sections 3–4]; [Q, Sections 5–8]; the citations of [R] and the main inequality; the constants and Table 1; an adversarial reading of the whole). It checked the cited statement numbers of [Q] by script and the quotations of [R] letter by letter, and recomputed Table 1 and the constants with several new programs (interval and exact rational arithmetic). It found no critical or major issue and 10 minor ones, all addressed: one displayed number rounded in

the wrong direction (34934.22 instead of 34934.23 in Section 6), a missing hypothesis $k \geq 2$ in the abstract, and points of notation, citation and wording.

- The eight cases of Table 1 were checked by the three programs named in Section 6; the second and the third were written by two of these AI reviewers, independently of the first and of each other.
- A numerical test on scaled analogues ($k = 14, 16$, inclinations $0.01\text{--}0.34$, $\delta \in \{1/20, 1/10\}$; 2,560 exactly verified closed packings, exact rational path tracing) found no counterexample to Lemmas 3.1–4.3 and to Theorem 5.1 with measured constants, and every negative control (removing one hypothesis or one term) made the corresponding check fail. Such a test cannot check the constants of [Q] and [R]. Its programs and results are in the folder `code/numerical_tests/`.

All reviewers are AI systems of the same family and may share blind spots. The argument rests on [Q, Sections 3–8] and on [R, Lemmas 3.1–3.5, 4.13 and 4.15]; Theorem 1.1 also rests on the computer-assisted [R, Lemma 4.10], whose box computations have not yet been reproduced by an independent implementation. These deserve priority in any human review.

Use of AI. Claude (Anthropic) was used extensively in developing the argument, writing the text and the programs, and in checking them. It is not an author; the author takes full responsibility for the contents.

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